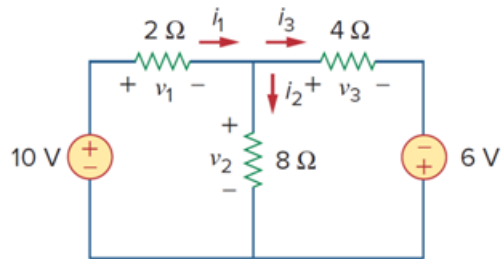


Problem

Find the currents and voltages in the circuit shown in Fig. 2.28.

Figure 2.28:



Step-by-step solution

Step 1 of 9

The circuit is shown in Figure 1.

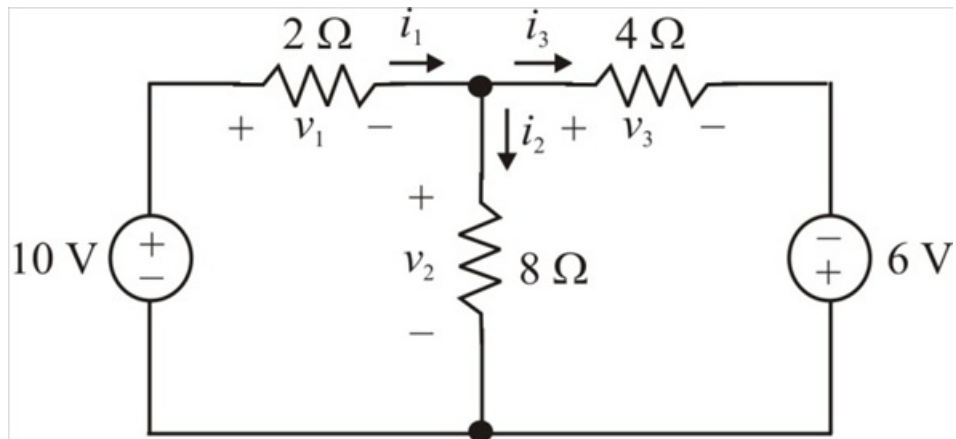


Figure 1

Step 2 of 9

The two loops and the node 'a' is shown in Figure 2.

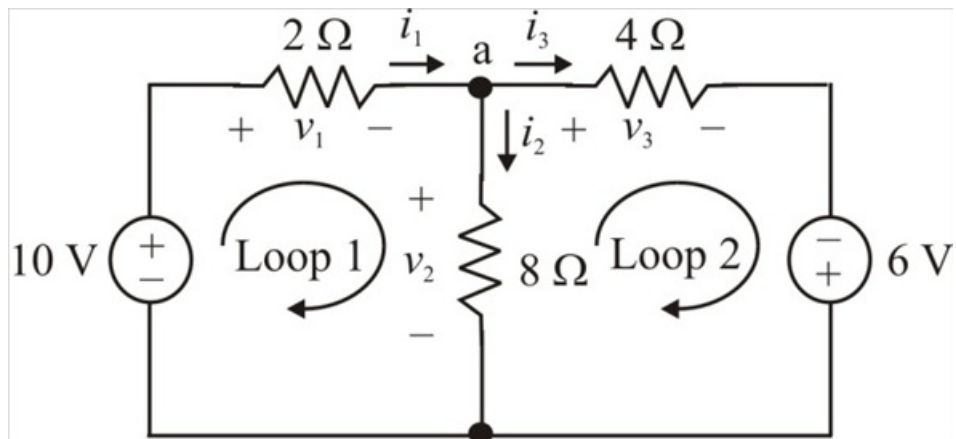


Figure 2

Step 3 of 9

Find i_o and v_o , apply Ohm's law and Kirchhoff's voltage law.

From Ohm's law,

$$v_1 = 2i_1 \dots\dots (1)$$

$$v_2 = 8i_2 \dots\dots (2)$$

$$v_3 = 4i_3 \dots\dots (3)$$

Apply Kirchhoff's current law to node 'a', we obtain,

$$i_1 = i_2 + i_3 \dots\dots (4)$$

Apply Kirchhoff's voltage law to loop1, we obtain,

$$-10 + v_1 + v_2 = 0$$

$$v_1 + v_2 = 10 \dots\dots (5)$$

Substituting equation (1) and (2) in equation (5), we get,

$$v_1 + v_2 = 10$$

$$2i_1 + 8i_2 = 10$$

$$i_1 = 5 - 4i_2$$

Therefore $i_1 = 5 - 4i_2 \dots\dots (6)$

Step 4 of 9

Apply Kirchhoff's voltage law to loop2, we obtain,

$$-6 - v_2 + v_3 = 0$$

$$v_3 = 6 + v_2 \dots\dots (7)$$

Substituting equation (2) and (3) in equation (7), we get,

$$v_3 = 6 + v_2$$

$$4i_3 = 6 + 8i_2$$

Therefore $4i_3 = 6 + 8i_2 \dots\dots (8)$

Step 5 of 9

Substitute the values of i_1 and i_3 in equation (4), we get,

$$i_1 = i_2 + i_3$$

$$5 - 4i_2 = i_2 + \frac{6 + 8i_2}{4}$$

$$20 - 20i_2 = 6 + 8i_2$$

$$28i_2 = 14$$

$$i_2 = 0.5 \text{ A}$$

Therefore, the current i_2 is 500mA.

Step 6 of 9

Substituting the values of i_2 in equation (6), we get,

$$\begin{aligned} i_1 &= 5 - 4i_2 \\ &= 5 - (4)(0.5) \\ &= 3 \text{ A} \end{aligned}$$

Therefore, the current i_1 is $\boxed{3 \text{ A}}$.

Step 7 of 9

Substituting the values of i_2 in equation (8), we get,

$$\begin{aligned} 4i_3 &= 6 + 8i_2 \\ &= 6 + (8)(0.5) \\ &= 10 \\ i_3 &= 2.5 \text{ A} \end{aligned}$$

Therefore, the current i_3 is $\boxed{2.5 \text{ A}}$.

Substitute the values of i_1 in equation (1), we get,

$$\begin{aligned} v_1 &= 2i_1 \\ &= (2)(3) \\ &= 6 \text{ V} \end{aligned}$$

Therefore, the voltage v_1 is $\boxed{6 \text{ V}}$.

Step 8 of 9

Substituting the values of i_2 in equation (2), we get,

$$\begin{aligned} v_2 &= 8i_2 \\ &= (8)(0.5) \\ &= 4 \text{ V} \end{aligned}$$

Therefore, the voltage v_2 is $\boxed{4 \text{ V}}$.

Step 9 of 9

Substituting the values of i_3 in equation (3), we get,

$$\begin{aligned} v_3 &= 4i_3 \\ &= (4)(2.5) \\ &= 10 \text{ V} \end{aligned}$$

Therefore, the voltage v_3 is $\boxed{10 \text{ V}}$.